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Re: Comprehensive Profiles of Human Cingulate Gyrus Effective Networks

Dear Dr. Takaki Komiyama,

It is with great pleasure that we are submitting our manuscript titled *Comprehensive Profiles of Human Cingulate Gyrus Effective Networks* for consideration for publication in *Science Advances*.

The networks of the human cingulum are studied most often due to their complexity, relevance to cognition and behavior, and role in various disease states. Recent clinical advances have enabled interventional methods to define and characterize such networks, specifically through direct electrical stimulation of brain regions. **The present work advances the field of human connectivity in three ways.** First, we provide a methodologically rigorous, multifaceted characterization of the human cingulum effective networks, acquired through direct stimulation of the cingulum subregions, which we believe will be an invaluable resource to any investigator researching the cingulum, its networks, or its function. Second, the results of the present study provide a breadth of new avenues for future research within these networks, which will lead to a deeper understanding of the human cingulum networks, their dynamics, and their functional roles. Finally, we provide a quantitative comparison of the measures used to define effective connectivity, evaluating their utility in classifying distinct effective networks. This is a topic often discussed in research on effective networks. It is often asserted that the current standards for evaluating effective connectivity are not sufficient, but to our knowledge, no other work has yet performed direct comparisons between measures of effective connectivity and their utility.

This work and its associated materials have not been published, nor are they under consideration for publication elsewhere. The authors note no direct conflicts of interest. Informed consent was obtained from all participants in this study after full explanations of the possible consequences. All processed data underlying the study will be uploaded to GitHub, along with the associated code. Raw data will be made available upon request, but will likely require a materials transfer agreement since it is human data collected in a clinical setting.

All authors have contributed to and approved the final submitted version of the manuscript. This manuscript is original work containing data that has not been previously published.

We thank you for considering our manuscript for publication in *Science Advances*.

Yours sincerely,

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(on behalf of the authors)